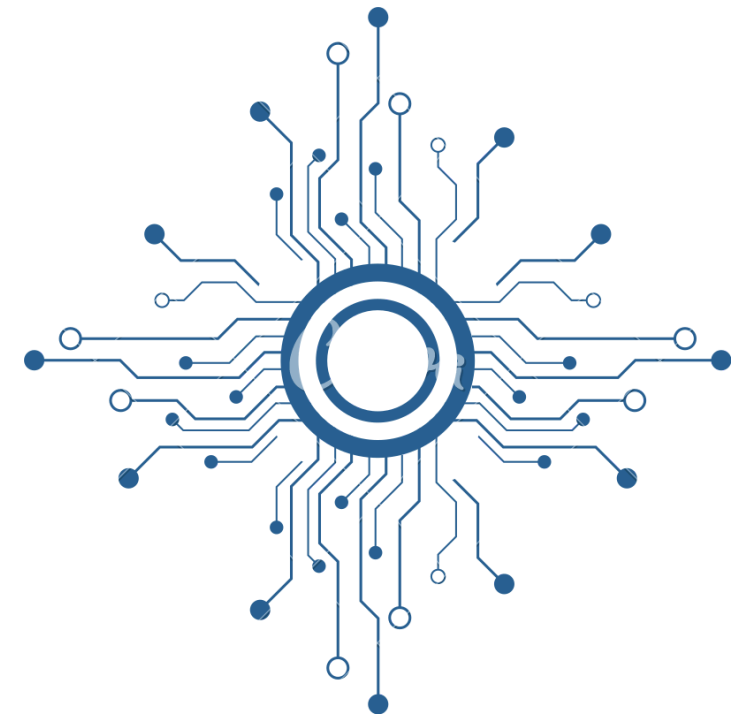
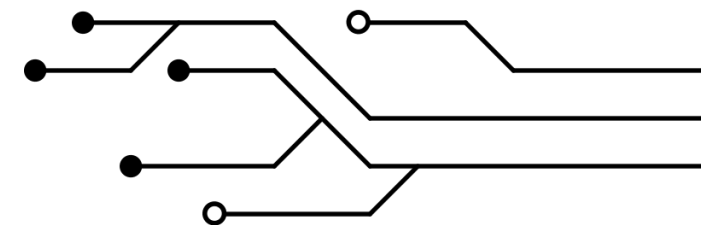




# CAR PRICE PREDICTION MODEL

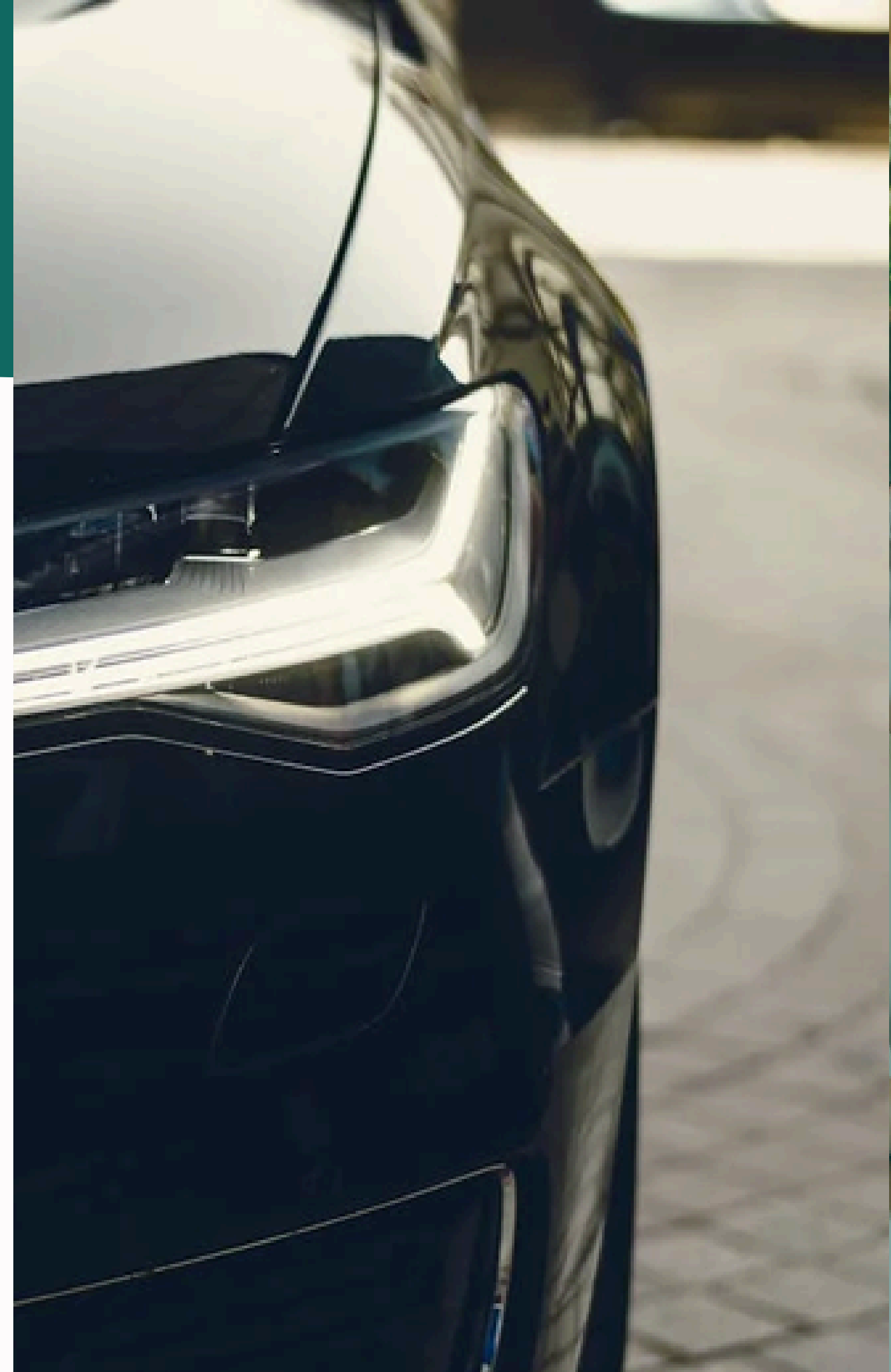
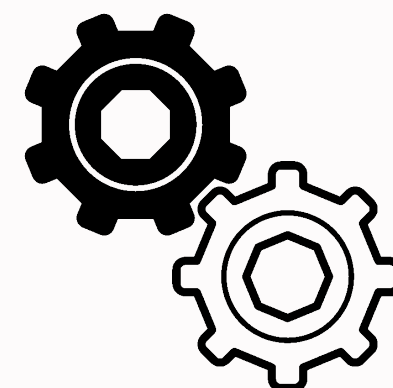
**MACHINE LEARNING CAPSTONE PROJECT**

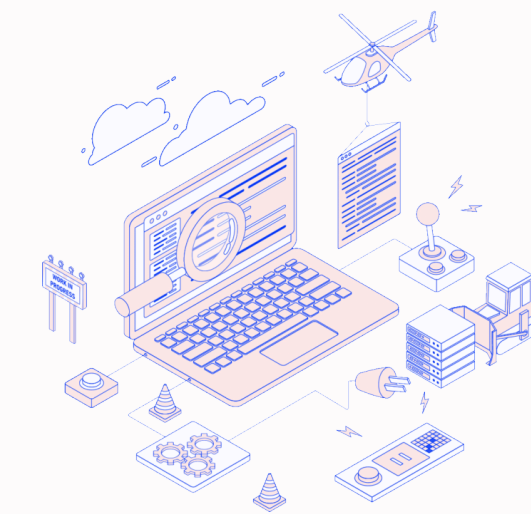
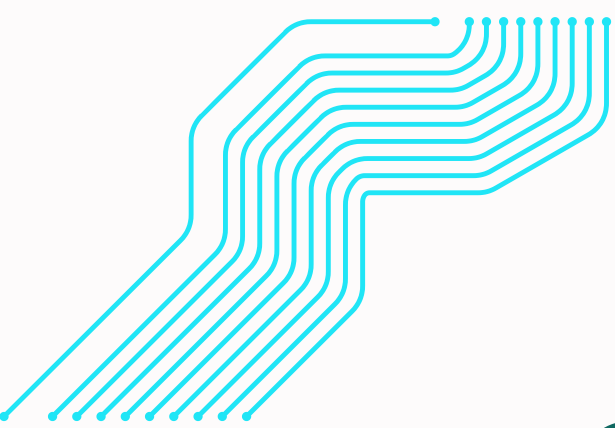
Anirban Paul  
Indian Institute of Technology, Jodhpur



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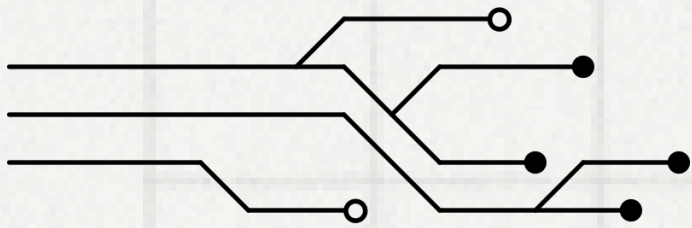




# OBJECTIVE

The Objective of this project is to build a machine learning model that accurately predicts car prices based on key features such as engine size, Horsepower, mileage, fuel cost, fuel type and others. The goal is to make a reliable tool for buyers and sellers to make informed decisions, uncover the most influential factors affecting car pricing, and deliver a scalable solution for the automotive market.





# DATASET OVERVIEW



## Dataset Source:

<https://www.kaggle.com/datasets/goyalshalini93/car-data>

## DATASET DICTIONARY

01

**Car\_ID:** Unique ID for each observation (Integer).  
**Symboling:** Insurance risk rating; +3 (risky) to -3 (safe) (Categorical).  
**Car Name:** Name of the car (Categorical).  
**Fuel Type:** Type of fuel used (gas/diesel) (Categorical).  
**Aspiration:** Type of aspiration in the car (Categorical).  
**Door Number:** Number of doors in the car (Categorical).  
**Car Body:** Type of car body (Categorical).  
**Drive Wheel:** Type of drive wheel (Categorical).  
**Engine Location:** Location of the car engine (Categorical).

02

**Wheelbase:** Distance between front and rear wheels (Numeric).  
**Car Length:** Length of the car (Numeric).  
**Car Width:** Width of the car (Numeric).  
**Car Height:** Height of the car (Numeric).  
**Curb Weight:** Weight of the car without passengers/baggage (Numeric).

03

**Engine Type:** Type of engine (Categorical).  
**Cylinder Number:** Number of cylinders in the car (Categorical).  
**Engine Size:** Size of the engine (Numeric).  
**Fuel System:** Type of fuel system (Categorical).  
**Bore Ratio:** Bore-to-stroke ratio (Numeric).  
**Stroke:** Stroke volume inside the engine (Numeric).

04

**Compression Ratio:** Compression ratio of the engine (Numeric).  
**Horsepower:** Engine power in horsepower (Numeric).  
**Peak RPM:** Maximum revolutions per minute (Numeric).  
**City MPG:** Mileage in the city (Numeric).  
**Highway MPG:** Mileage on the highway (Numeric).  
**Price:** Dependent variable, price of the car (Numeric).

## DATA CLEANING

### MISSING DATA

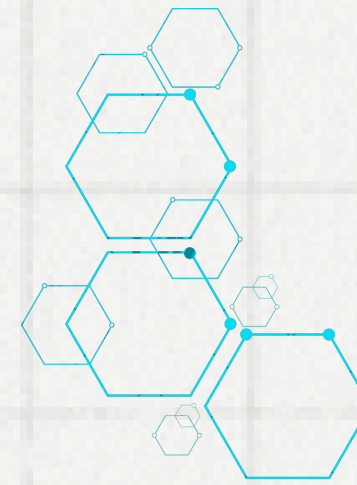
Missing data was identified and addressed by filling with median values to ensure that no null values would disrupt the modeling process.

### DUPLICATE DATA

Duplicated rows in the dataset were checked and removed to avoid bias and redundancy during analysis.

### FEATURE TRANSFORMATION

New features were engineered to enhance the dataset. For example company name is obtained from car name



# DATA PREPROCESSING

## FEATURE SCALING

### STANDARTSCALLER

Scaling was performed on numeric variables (e.g., Horsepower, Curb Weight, Wheelbase) to ensure all features have a comparable range, which is critical for machine learning models sensitive to the magnitude of input data (e.g., gradient-based methods).

### DATA SPLITTING

Data splitting refers to dividing the dataset into training and testing sets. Typically, an 80:20 split was used to ensure the model learns from the training set and is validated on unseen test data.

### NORMALIZATION

Makes sure that feature values of each point (each row) sum up to 1 (L1 norm)  
Useful for count data (e.g. word counts in documents)  
Can also be used with L2 norm (sum of squares is 1)

- Useful when computing distances in high dimensions
- Normalized Euclidean distance is equivalent to cosine similarity

# MODELLING

## BASELINE MODEL

A simple model used to set a performance benchmark.

This model helps evaluate the improvement brought by more complex algorithms.

Algorithms used :  
Linear Regression  
Decision Tree

## LINEAR REGRESSION

A linear regression is a linear approximation of a casual relationship between two or more variables. They are highly valueable as they are highly used in prediction and analysis of scattered data.

$$\hat{y} = \beta_0 + \beta_1 x$$

## DECISION TREE

- Decision Tree is a Supervised learning technique that can be used for both classification and Regression problems, but mostly it is preferred for solving Classification problems. It is a tree-structured classifier, where internal nodes represent the features of a dataset, branches represent the decision rules and each leaf node represents the outcome.

## ENSEMBLE MODEL

Combines predictions from multiple models to improve performance and robustness.

Algorithms used :  
Random Forest  
Gradient Descent/Boosting

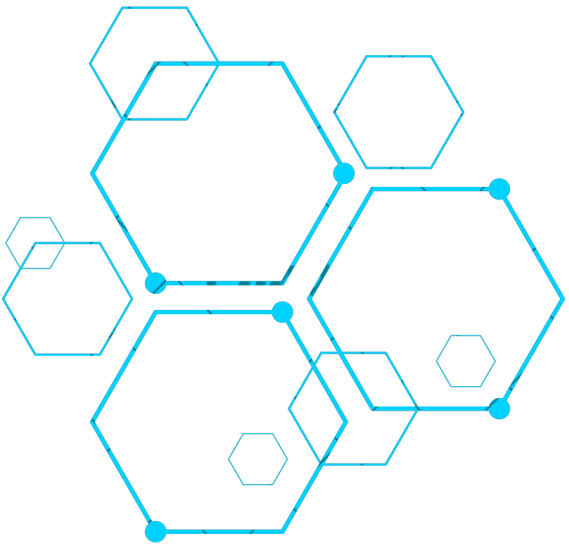
## RANDOM FOREST

Random Forest is a classifier that contains a number of decision trees on various subsets of the given dataset and takes the average to improve the predictive accuracy of that dataset. Instead of relying on one decision tree, the random forest takes the prediction from each tree and based on the majority votes of predictions, and it predicts the final output.

The greater number of trees in the forest leads to higher accuracy and prevents the problem of overfitting.

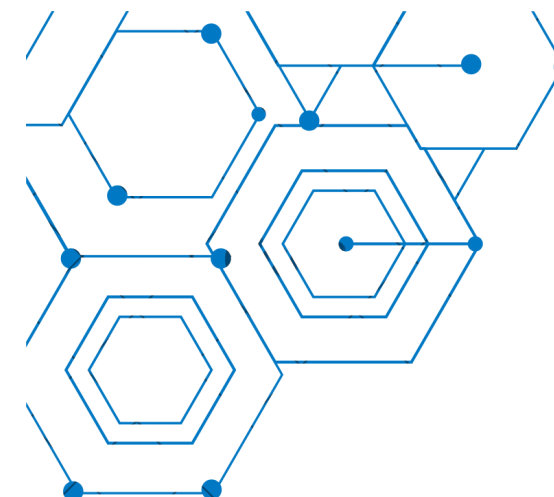
## GRADIENT DESCENT

Gradient Descent is defined as one of the most commonly used iterative optimization algorithms of machine learning to train the machine learning and deep learning models. It helps in finding the local minimum of a function.



# KEY METRICS:

- 01**  $R^2$  Score
- 02** Mean Squared Error (MSE)
- 03** Mean Absolute Error (MAE)
- 04** Root Mean Squared Error



# EVALUATION

**Model evaluation involves analyzing how well a trained model performs when tested on new data. Various evaluation metrics help assess its performance. These include accuracy, precision, recall, the F1 score, cross-validation and AUC-ROC.**



# MODEL PERFORMANCE COMPARISION TABLE

| Model             | Mean Squared Error (MSE) | Mean Absolute Error (MAE) | R-squared (R²) | Root Mean Squared Error (RMSE) |
|-------------------|--------------------------|---------------------------|----------------|--------------------------------|
| Random Forest     | 3,275,513.57             | 1,284.36                  | 0.9586         | 1,888.46                       |
| Gradient Boosting | 5,688,260.73             | 1,684.46                  | 0.9279         | 2,385.01                       |
| Linear Regression | 8,635,315.60             | 1,924.64                  | 0.8906         | 2,938.59                       |
| Decision Tree     | 8,920,432.77             | 1,955.96                  | 0.8870         | 2,986.71                       |

# KEY INSIGHT :



## Best Performance

The Random Forest model has the lowest MSE and RMSE, indicating it is the most accurate model for predicting car prices.



## Comparision of Ensemble Models

Both ensemble models (Random Forest and Gradient Boosting) outperform the baseline models (Linear Regression and Decision Tree) in all metrics

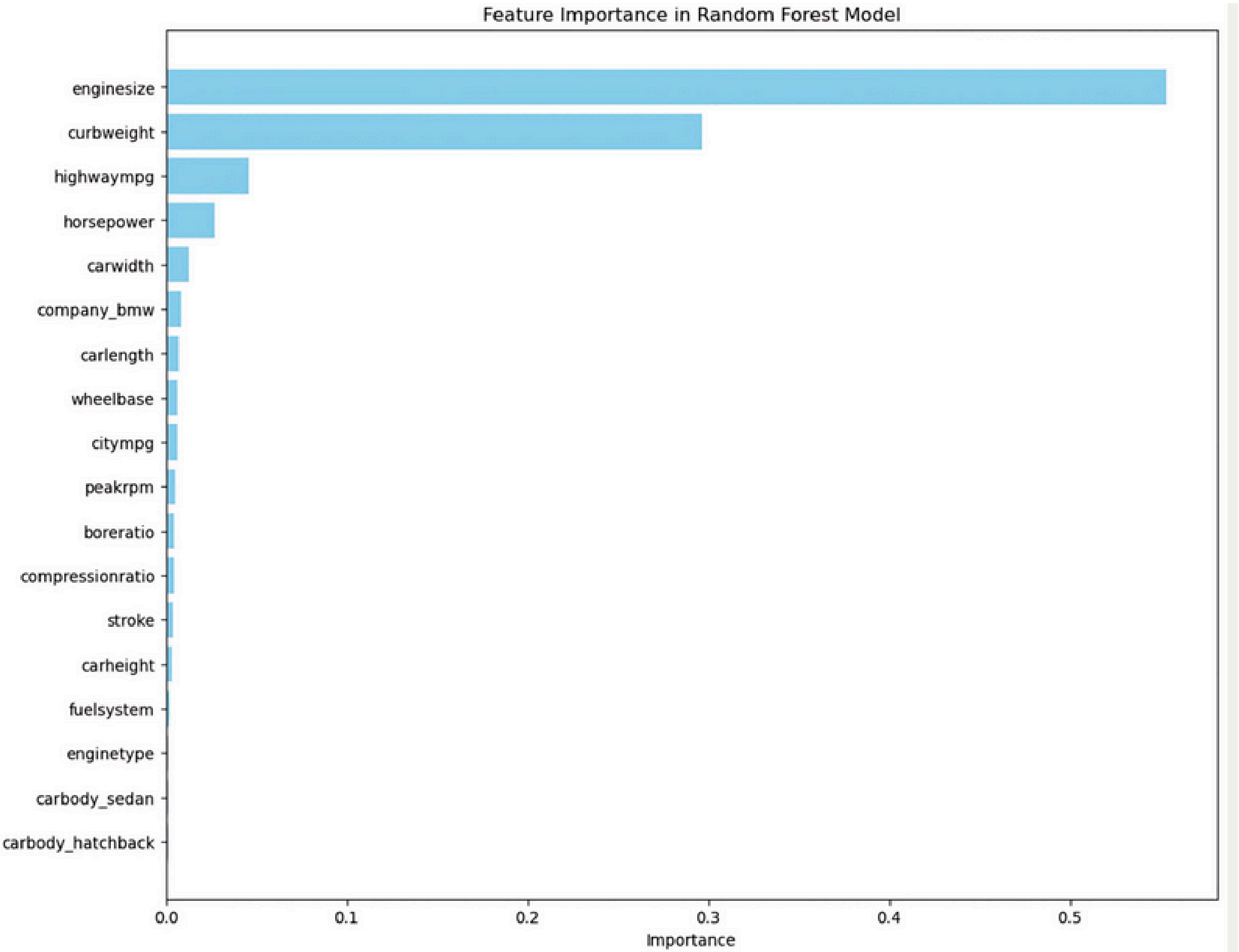
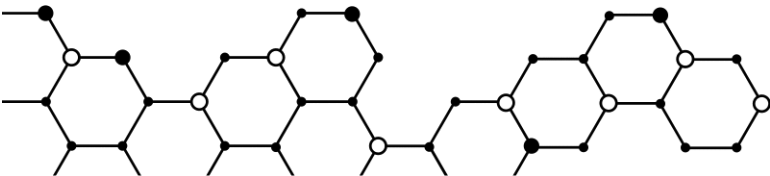


## Baseline model limitations

The baseline models show significantly higher error metrics, suggesting they are less effective in capturing the underlying patterns in the data.

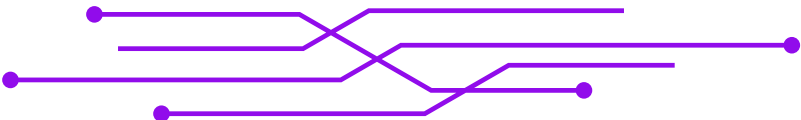


# FEATURE IMPORTANT



## KEY INSIGHT :

- i) Engine size is the most significant factor affecting car price predictions, indicating that engine capacity plays a primary role in determining car value
- ii) Curb weight is the second most important feature, suggesting that the vehicle's weight (likely linked to material quality and performance) is crucial.
- iii) Highway MPG and Horsepower also stands out as significant contributors, highlighting that fuel efficiency and vehicle power are key determinants of car pricing.



# Training and Testing Data (An Overview of Orange Software)



# CONCLUSION

## .....• MODEL ACCURACY

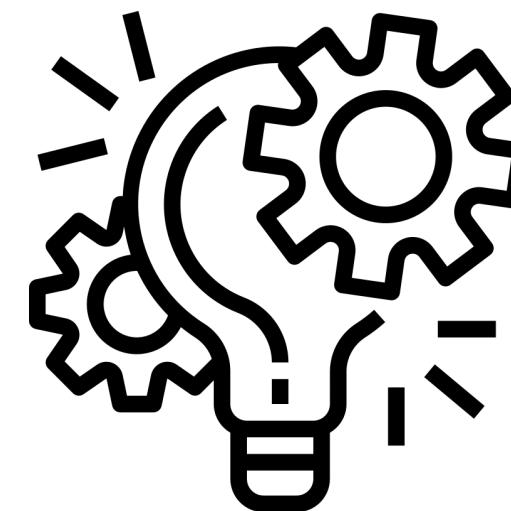
The Random Forest model demonstrated superior performance with an  $R^2$  score of 0.9586, making it a highly accurate tool for predicting car prices.

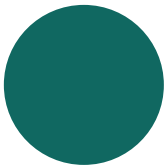
## .....• KEY INFLUENTIAL FACTORS

Engine size and curb weight emerged as the most important predictors, followed by highway MPG and horsepower, highlighting the critical role of performance and efficiency in determining car prices.

## .....• SCALABILITY AND PRACTICALITY

The model is scalable and adaptable to various automotive markets, providing a reliable framework for similar predictions. Its insights can guide manufacturers, buyers, and sellers in making informed decision





**THANK YOU**